

Hackathon Problem Statements

Regd No: 25091A32J6, 25091A3257

Problem Statement No: 65

Peer Tutoring Matchmaking System Design a peer tutoring matchmaking system using graph-based compatibility scoring that pairs students needing help in specific subjects with peer tutors strong in those subjects, factoring in schedule availability. Use OOP to model students, subjects, and matching sessions. This creates a low-cost, scalable academic support system, especially useful in resource-constrained institutions lacking sufficient faculty for remedial classes.

Regd No: 25091A3285, 25091A3286

Problem Statement No: 68

Attendance Anomaly Detector Build an anomaly detection system that analyzes biometric/log-based attendance patterns using statistical outlier detection to flag suspicious patterns like proxy attendance or unusual check-in/check-out sequences. Use OOP to model attendance records, students, and anomaly flags. This helps institutions maintain attendance integrity, particularly relevant for compliance with regulatory attendance requirements.

Regd No: 25091A32M8, 25091A32N8

Problem Statement No: 82

Energy Consumption Anomaly Detector Create a household energy anomaly detection system using clustering and outlier detection on simulated smart-meter consumption data to flag unusual usage spikes that may indicate faulty appliances, theft, or wastage. Use OOP to model households, meters, and consumption records. This supports utility companies and consumers in identifying inefficiencies and reducing unnecessary electricity costs.

Regd No: 25091A3218, 25091A3216

Problem Statement No: 111

Disaster Relief Supply Chain Optimizer Design a relief supply distribution system using network flow algorithms to optimize the movement of essential supplies (food, water, medicine) from central depots to multiple disaster-affected distribution points with capacity and route constraints. Use OOP to model depots, distribution points, and supply inventories. This ensures faster, more equitable relief distribution during large-scale disasters.

Regd No: 25091A3294, 25091A3239

Problem Statement No: 141

Phishing URL Detection System Build a phishing URL classifier using classification models (logistic regression/ decision trees) trained on URL structural features (length, special characters, domain age proxies) from public phishing datasets to flag suspicious links in real time. Use OOP to model URLs, features, and classification results. This provides a practical, lightweight defense layer against a widespread and constantly evolving cyber threat.

Regd No: 25091A32F5, 25091A32H1

Problem Statement No: 142

Password Strength Analyzer & Breach-Checker Design a password security tool using hashing algorithms to check user passwords against known breach datasets without ever storing or transmitting the plaintext password, alongside a strength-scoring engine that evaluates entropy and common-pattern usage. Use OOP to model password policies and scoring rules. This promotes better personal cybersecurity hygiene through practical, privacy-respecting tooling.